ViRGo: Virtual Role-Graph Embedding for Structural Identity

Research proposal

1. Motivation

I am currently reproducing the Identity2Vec (I2V) results. Producing the embedding for a single dataset (webkb, 265 nodes) took approximately 28 hours, because eigenvector centrality and the degree distribution are recomputed inside the walk loop at every step, despite being invariant over the run. This confirms the efficiency concern raised in our meeting and motivates two distinct changes.

2. Proposed Changes

2.1 Fix 1 — Precompute-and-cache the structural signal

Compute the graph-level degree and eigenvector centrality once per graph (both are computed on the static full graph in I2V, (self.G — the full, unchanging graph) so caching is exact, store them, and have the Poisson/KL scoring read from the cache rather than recomputing per step. This removes the redundant per-iteration centrality computation without altering the mathematics, and is the prerequisite for any scalable variant.

2.2 Fix 2 — Replace walk + Skipgram with GNN message passing on a virtual graph

We will keep the I2V front end (degree/eigenvector distributions → KL-divergence λ → Poisson Ψ). Instead of sampling guided walks and learning with Skipgram, We will materialize a *virtual structural-similarity graph* by connecting each node to its top-K most structurally similar nodes under Ψ, then learn embeddings with an inductive GNN encoder like GraphSAGE, with GIN as a more expressive alternative and GAT as an ablation. Message passing then aggregates over role-similar nodes rather than physical neighbors, so structurally equivalent but distant nodes inform each other directly, the mesoscopic property I2V targets, now enforced architecturally.

3. Primary Objective and Downstream Tasks

The contribution is the *encoder*, not the task: can a modern GNN over the Poisson/KL virtual graph improve structural-identity embeddings versus walk + Skipgram? We will retain link prediction and node classification as in I2V for direct comparability, then extend to **graph anomaly detection**, where the virtual graph is well-suited: structurally anomalous nodes fall far from role-consistent clusters, so role-aware message passing should sharpen separability. This feeds our COBench effort (Cyril) on collaborative semi-supervised detection: cleaner role-based embeddings should yield more reliable cross-model **pseudo-labels** on unknown labels, improving AUC/AP. As a secondary motivation, the same compact structural embeddings may also be explorable as a graph-summarization signal to help mitigate LLM context-window limits when feeding large-graph structure to an LLM.

4. Status and Next Steps

Reproduction of I2V is in progress, with Cora and Citeseer as the comparison subset. On approval, I will:

1. Implement Fix 1 and validate that embeddings results match the baseline (I2V).
2. Build the virtual-graph + GraphSAGE/GIN pipeline (K treated as a tuned hyperparameter).
3. Benchmark link prediction and node classification against I2V for direct comparability;
4. Then evaluate ViRGo embeddings on standard graph anomaly detection (GADBench / PyGOD)